

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 3: Classification

Ch 3 – Classification – Numerical Problems & Solutions

6 distinct methods · 12 PYQ instances solved · every answer independently recomputed. Decision-tree and Naive-Bayes figures cross-checked against Han & Kamber 3rd ed Examples 8.1–8.4, which use the same dataset the IOE papers copy.

Methods

- **M1** Decision tree by information gain (ID3) 79 Bh, 78 Bh, 76 Ch
- **M2** Gini index 75 Ch, 71 Ch
- **M3** Naive Bayes classification 72 Ch, 79 Bh
- **M4** k -Nearest Neighbour 80 Bh
- **M5** Confusion matrix metrics **8 papers**
- **M6** ANN gradients (backpropagation) 76 Ch

M1. Decision tree by information gain (ID3)

Method

1. Compute $Info(D) = -\sum p_i \log_2 p_i$ for the whole set.
 2. For each attribute A : split D , compute $Info_A(D) = \sum \frac{|D_j|}{|D|} Info(D_j)$, then $Gain(A) = Info(D) - Info_A(D)$.
 3. **Highest gain wins** → becomes the node's test.
 4. Recurse on each branch. Stop when a branch is **pure** ($Info = 0$).
- **Time-savers**: a pure partition contributes 0; a 50/50 partition contributes exactly 1; two partitions with the same class ratio have the same $Info$, so compute once.
 - $\log_2 x = \ln x / \ln 2$. Keep 4 decimals until the end.

Problem 1.1 10

QUESTION AS PRINTED 2079 Bhadra · Q3 · [10]

Draw decision tree for the given data using ID3 algorithm.

Age	Income	Student	Credit_Rating	Buy's_Computer
Youth	High	No	Fair	No
Youth	High	No	Excellent	No
Middle_Aged	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes
Senior	Low	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No
Middle_Aged	Low	Yes	Excellent	Yes
Youth	Medium	No	Fair	No
Youth	Low	Yes	Fair	Yes
Senior	Medium	Yes	Fair	Yes
Youth	Medium	Yes	Excellent	Yes
Middle_Aged	Medium	No	Excellent	Yes
Middle_Aged	High	Yes	Fair	Yes
Senior	Medium	No	Excellent	No

Given

14 tuples. Class counts: **Yes = 9, No = 5**.

Step 1: entropy of the whole set

$$Info(D) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.4098 + 0.5305 = \mathbf{0.940}$$

Step 2: gain of each attribute

Attribute	Partition (Yes, No)	$Info_A(D)$	Gain
Age	Youth (2,3) · Middle_Aged (4,0) · Senior (3,2)	$\frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971) = 0.694$	0.246
Income	High (2,2) · Medium (4,2) · Low (3,1)	$\frac{4}{14}(1.0) + \frac{6}{14}(0.918) + \frac{4}{14}(0.811) = 0.911$	0.029
Student	Yes (6,1) · No (3,4)	$\frac{7}{14}(0.592) + \frac{7}{14}(0.985) = 0.788$	0.151
Credit_Rating	Fair (6,2) · Excellent (3,3)	$\frac{8}{14}(0.811) + \frac{6}{14}(1.0) = 0.892$	0.048

- **Age has the highest gain → Age is the root.**
- Note Middle_Aged is (4, 0): **pure**, so that branch is immediately a leaf “Yes”. That purity is what makes Age win.

Step 3: recurse on the Youth branch

(5 tuples: 2 Yes, 3 No, $Info = 0.971$)

- Student = Yes → (2 Yes, 0 No) Student = No → (0 Yes, 3 No)
- Both pure → $Info_{Student} = 0$ → $Gain = 0.971$, the maximum possible.
- **Student** is the test under Youth.

Step 4: recurse on the Senior branch

(5 tuples: 3 Yes, 2 No, $Info = 0.971$)

- Credit = Fair → (3 Yes, 0 No) Credit = Excellent → (0 Yes, 2 No)
- Both pure → $Gain = 0.971$. **Credit_Rating** is the test under Senior.

Final tree

- **Age?**
 - Youth → **Student?**
 - no → **No**
 - yes → **Yes**
 - Middle_Aged → **Yes**
 - Senior → **Credit_Rating?**
 - fair → **Yes**
 - excellent → **No**
- Tree has **5 leaves**, depth 2, and classifies all 14 training tuples correctly.

8.2 Decision Tree Induction 331

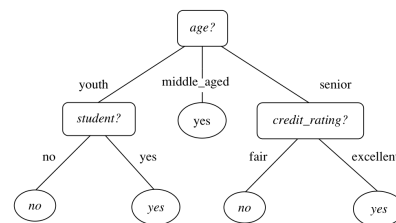


Figure 8.2 A decision tree for the concept *buys.computer*, indicating whether an *AlIElectronics* customer is likely to purchase a computer. Each internal (nonleaf) node represents a test on

The same tree as printed in Han & Kamber 3rd ed, Fig 8.2. Our independently computed gains match their Example 8.1 exactly.

Problem 1.2 8

QUESTION AS PRINTED 2078 Bhadra · Q4 · [8]

Construct a decision tree for the following data set using information gain.

Predict the class label for a data point with values <Female, 2, standard, high>

Gender	Car ownership	Travel cost	Income level	Transport mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

Given

Classes: Bus = 4, Car = 3, Train = 3. $n = 10$.

$$Info(D) = -\frac{4}{10} \log_2 \frac{4}{10} - \frac{3}{10} \log_2 \frac{3}{10} - \frac{3}{10} \log_2 \frac{3}{10} = 0.5288 + 0.5211 + 0.5211 = \mathbf{1.571}$$

Travel cost

Cheap (4 Bus, 0 Car, 1 Train) · Expensive (0,3,0) · Standard (0,0,2)

- $Info(\text{Cheap}) = -\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5} = 0.722$
- Expensive and Standard are both **pure** $\rightarrow Info = 0$
- $Info_{TravelCost}(D) = \frac{5}{10}(0.722) + \frac{3}{10}(0) + \frac{2}{10}(0) = 0.361$
- $Gain = 1.571 - 0.361 = \mathbf{1.210}$
- Every other attribute scores far lower (Gender 0.125, Car ownership 0.534, Income level 0.971), so **Travel cost is the root** – two of its three branches are pure.
- Only the *Cheap* branch needs splitting: 5 tuples (4 Bus, 1 Train). Splitting it on **Gender** gives Male (3 Bus, 0 Train) and Female (1 Bus, 1 Train); splitting on **Car ownership** gives 0 (2 Bus), 1 (2 Bus, 1 Train). Neither is perfect; Gender has the higher gain, then Car ownership finishes the Female sub-branch.

Prediction for <Female, 2, standard, high>

- Travel cost = **standard** $\rightarrow$ that branch is **pure Train**.
- **No further tests are needed.** Answer: **Train**.

M2. Gini index

Method

$$Gini(D) = 1 - \sum_i p_i^2 \quad Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2) \quad \Delta Gini = Gini(D) - Gini_A(D)$$

- **Gini uses BINARY splits only.** For an attribute with v values you must try all $2^{v-1} - 1$ ways of splitting the values into 2 subsets and keep the best.
- Pick the largest $\Delta Gini$ (equivalently the smallest $Gini_A$).

Worked example (same 14-tuple dataset) 6

QUESTION AS PRINTED 2075 Chaitra · Q3 · [6+4]

Define Decision Tree Classifier with Gini-Index with suitable example. How can you handle overfitting in Decision Tree?

QUESTION AS PRINTED 2071 Chaitra · Q3 · [10]

What is a decision tree? Explain Gini Index with suitable example.

Neither paper prints a dataset – the “suitable example” is yours to supply. The 14-tuple *buys_computer* set below is the safe one to reuse. verified/added

- $Gini(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 1 - 0.4133 - 0.1276 = \mathbf{0.459}$

- Split **Income** as {low, medium} vs {high}:
 - $D_1 = 10$ tuples (7 Yes, 3 No): $Gini = 1 - (0.7)^2 - (0.3)^2 = 0.420$
 - $D_2 = 4$ tuples (2 Yes, 2 No): $Gini = 1 - (0.5)^2 - (0.5)^2 = 0.500$
 - $Gini_{Income}(D) = \frac{10}{14}(0.420) + \frac{4}{14}(0.500) = 0.300 + 0.143 = 0.443$
 - $\Delta Gini = 0.459 - 0.443 = \mathbf{0.016}$
- Split **Age** as {youth, senior} vs {middle_aged}:
 - $D_1 = 10$ tuples (5 Yes, 5 No): $Gini = 0.500$
 - $D_2 = 4$ tuples (4 Yes, 0 No): $Gini = \mathbf{0}$ (pure)
 - $Gini_{Age}(D) = \frac{10}{14}(0.500) + \frac{4}{14}(0) = 0.357$
 - $\Delta Gini = 0.459 - 0.357 = \mathbf{0.102} \rightarrow \text{best, so Age is chosen}$
- Same winner as information gain, which is the usual (not guaranteed) outcome.

M3. Naive Bayes classification

Method

1. Priors: $P(C_i) = |C_{i,D}|/|D|$.
 2. For each attribute value in X : $P(x_k|C_i)$ from counts within class C_i .
 3. $P(X|C_i) = \prod_k P(x_k|C_i)$.
 4. Compare $P(X|C_i)P(C_i)$ across classes; **largest wins**.
- **Never divide by $P(X)$** – it is identical for every class and cancels.
 - Any zero count $\rightarrow$ apply **Laplacian correction** $\frac{n_k+1}{|C_{i,D}|+k}$.
 - Lay the work out as a small table of conditional probabilities; it earns method marks even if arithmetic slips.

Problem 3.1 10

QUESTION AS PRINTED 2072 Chaitra · Q3 · [10]

Predict Class label using naive Bayesian classifier for $X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit-rating} = \text{fair})$ using the following data set.

RID	Age	Income	Student	Credit-rating	Class Buy computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle-age	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle-age	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle-age	Medium	No	Excellent	Yes
13	Middle-age	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

Given

14 tuples. $P(\text{Yes}) = 9/14 = 0.643$, $P(\text{No}) = 5/14 = 0.357$.

Conditional probabilities

	Yes	No
$P(\text{age} = \text{youth})$	$2/9 = 0.222$	$3/5 = 0.600$
$P(\text{income} = \text{med})$	$4/9 = 0.444$	$2/5 = 0.400$
$P(\text{student} = \text{yes})$	$6/9 = 0.667$	$1/5 = 0.200$
$P(\text{credit} = \text{fair})$	$6/9 = 0.667$	$2/5 = 0.400$

Combine

- $P(X|\text{Yes}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = \mathbf{0.0439}$
- $P(X|\text{No}) = 0.600 \times 0.400 \times 0.200 \times 0.400 = \mathbf{0.0192}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.0439 \times 0.643 = \mathbf{0.028}$
- $P(X|\text{No})P(\text{No}) = 0.0192 \times 0.357 = \mathbf{0.007}$

Final answer

$0.028 > 0.007 \rightarrow \text{buys_computer} = \text{Yes}.$

- Matches Han & Kamber 3rd ed Example 8.4.

Problem 3.2 6

QUESTION AS PRINTED 2079 Bhadra · Q4 · [6]

Suppose you have a test record “ $X = (\text{Home Owner} = \text{No}, \text{Marital Status} = \text{Married}, \text{Income} = \$120\text{K})$ ”. Your job is to classify this record using Naive Bayesian Classification. Use the following table for your calculations.

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Given

Defaulted: **No = 7, Yes = 3**. $P(\text{No}) = 0.7$, $P(\text{Yes}) = 0.3$.

Step 1: categorical attributes

- $P(\text{HO} = \text{No} | \text{No}) = 4/7 = 0.571$ $P(\text{HO} = \text{No} | \text{Yes}) = 3/3 = 1.0$
- $P(\text{Married} | \text{No}) = 4/7 = 0.571$ $P(\text{Married} | \text{Yes}) = 0/3 = 0 \leftarrow \text{zero!}$
- Apply **Laplacian correction** (Marital Status has $k = 3$ values): $P(\text{Married} | \text{Yes}) = \frac{0+1}{3+3} = 0.167$; and $P(\text{HO} = \text{No} | \text{Yes}) = \frac{3+1}{3+2} = 0.8$.

Step 2: continuous attribute (Income), Gaussian

$$g(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Class **No**: incomes 125, 100, 70, 120, 60, 220, 75 $\rightarrow \mu = 110, \sigma = 54.54$

$$g(120) = \frac{1}{\sqrt{2\pi}(54.54)} e^{-\frac{(10)^2}{2(2975)}} = 0.00719$$

- Class **Yes**: incomes 95, 85, 90 $\rightarrow \mu = 90, \sigma = 5$

$$g(120) = \frac{1}{\sqrt{2\pi}(5)} e^{-\frac{(30)^2}{2(25)}} = 0.0798 e^{-18} \approx 1.2 \times 10^{-9}$$

- **The Yes class has a tiny spread ($\sigma = 5$) around \$90K, so \$120K is 6 standard deviations out** — effectively impossible under that model.

Step 3: combine

- $P(X|\text{No})P(\text{No}) = 0.571 \times 0.571 \times 0.00719 \times 0.7 = \mathbf{1.64 \times 10^{-3}}$
- $P(X|\text{Yes})P(\text{Yes}) = 0.8 \times 0.167 \times 1.2 \times 10^{-9} \times 0.3 \approx \mathbf{4.8 \times 10^{-11}}$

Final answer

Defaulted Borrower = No (larger by roughly 10^7).

M4. k -Nearest Neighbour

Method

1. Compute the Euclidean distance from the query to **every** training tuple.
 2. Sort ascending, take the k smallest.
 3. **Majority class** among those k is the prediction.
- **Show the full distance table** — that is where the marks are, not the final label.
 - You may compare squared distances to save time, but write the roots in the final table.

Problem 4.1 6

QUESTION AS PRINTED 2080 Bhadra · Q2 (b) · [3+6]

When do we use classifier? You have the following information about the flower. Your job is to classify the given flower with SepalLength = 7, SepalWidth = 3.2, PetalLength = 4.7, and PetalWidth = 1.4. Use KNN algorithm for $K = 3$ with Euclidean distance matrix.

Id	Sepal_Lenth	Spal_Width	Petal_Lenth	Petal_Width	Labal
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	6	2.2	4	1	versicolor
5	6.1	2.9	4.7	1.4	versicolor
6	5.6	2.9	3.6	1.3	versicolor
7	6.7	3.1	4.4	1.4	versicolor

Column names are

the paper's own spellings.

Id	Tuple	Squared differences	d	Label
1	(5.1, 3.5, 1.4, 0.2)	$3.61 + 0.09 + 10.89 + 1.44 = 16.03$	4.004	setosa
2	(4.9, 3.0, 1.4, 0.2)	$4.41 + 0.04 + 10.89 + 1.44 = 16.78$	4.096	setosa
3	(4.7, 3.2, 1.3, 0.2)	$5.29 + 0 + 11.56 + 1.44 = 18.29$	4.277	setosa
4	(6.0, 2.2, 4.0, 1.0)	$1 + 1 + 0.49 + 0.16 = 2.65$	1.628	versicolor
5	(6.1, 2.9, 4.7, 1.4)	$0.81 + 0.09 + 0 + 0 = 0.90$	0.949	versicolor
6	(5.6, 2.9, 3.6, 1.3)	$1.96 + 0.09 + 1.21 + 0.01 = 3.27$	1.808	versicolor
7	(6.7, 3.1, 4.4, 1.4)	$0.09 + 0.01 + 0.09 + 0 = 0.19$	0.436	versicolor

Step: pick the 3 nearest

- Sorted: $d_7 = 0.436 < d_5 = 0.949 < d_4 = 1.628 < d_6 = 1.808 < d_1 = 4.004 \dots$
- The 3 nearest are **Id 7, Id 5, Id 4** → versicolor, versicolor, versicolor.

Final answer

versicolor (unanimous, 3 of 3).

- Note all 3 setosa tuples sit at distance > 4 : they have tiny petals, so they are nowhere near the query. The vote was never in doubt.

M5. Confusion matrix metrics (8 papers — highest-yield numerical in the syllabus)

DO THIS FIRST, EVERY TIME. Copy the row and column headings onto your answer sheet and mark which axis is **actual**. The IOE papers **flip the orientation between years**.

Proof: 75 Ash and 72 Ch print the *identical* four numbers 21, 6, 7, 41 — but 75 Ash has actual on the rows and 72 Ch has predicted on the rows. The result is that **sensitivity and precision swap places** (0.778 / 0.750 becomes 0.750 / 0.778). Same numbers, different answers. verified/added

Method

- Identify TP , FN , FP , TN from the orientation. Then $P = TP + FN$ (all actual positives), $N = FP + TN$ (all actual negatives).
- Accuracy = $\frac{TP+TN}{P+N}$ Error = 1–accuracy
- Sensitivity = recall = $TPR = \frac{TP}{P}$ Specificity = $TNR = \frac{TN}{N}$
- FPR = false alarm rate = $\frac{FP}{N} = 1 - \text{specificity}$ Precision = $\frac{TP}{TP+FP}$

Problem 5.1 5

QUESTION AS PRINTED 2074 Chaitra · Q3 · [2+3+5]

Draw clear block diagram depicting different stages in classification. Explain the inverse relation between precision and recall. Given the confusion matrix, determine accuracy, sensitivity and precision of the classifier model.

Actual \ Predicted	Positive	Negative
Positive	142	40
Negative	98	720

"Positive" is the paper's own spelling.

- $TP = 142, FN = 40, FP = 98, TN = 720 \rightarrow P = 182, N = 818, \text{total} = 1000$
- Accuracy = $\frac{142+720}{1000} = \mathbf{0.862}$
- Sensitivity = $\frac{142}{182} = \mathbf{0.780}$
- Precision = $\frac{142}{142+98} = \frac{142}{240} = \mathbf{0.592}$
- Specificity = $\frac{720}{818} = \mathbf{0.880}$

Problem 5.2 4

QUESTION AS PRINTED 2078 Bhadra · Q6 · [4]

Calculate: Accuracy, TPR, FPR and Precision for the given confusion matrix for a classifier.

		Actual Class	
		Class 1	Class 2
Predicted Class	Class 1	142	40
	Class 2	98	720

Same four numbers as 5.1, but the rows are Predicted here.

- Rows are predicted $\rightarrow TP = 142, FP = 40, FN = 98, TN = 720$
- $P = 142 + 98 = 240, N = 40 + 720 = 760$
- Accuracy = $\frac{142+720}{1000} = \mathbf{0.862}$ (unchanged – the diagonal is the same)
- $TPR = \frac{142}{240} = \mathbf{0.592}$ $FPR = \frac{40}{760} = \mathbf{0.053}$
- Precision = $\frac{142}{142+40} = \frac{142}{182} = \mathbf{0.780}$
- **Compare with 5.1:** sensitivity and precision have exchanged values. This is the orientation trap, live.

Problems 5.3 and 5.4 10

QUESTION AS PRINTED 2075 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	21	6
	Class 2	7	41

Calculate: Accuracy, Sensitivity, Specificity and Precision.

QUESTION AS PRINTED 2072 Chaitra · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		actual class	
		class1	class2
predicted class	class1	21	6
	class2	7	41

Calculate a. accuracy b. sensitivity c. specificity d. precision e. recall

Same four numbers, opposite orientation – that is the whole point of solving them side by side.

75 Ash – rows are Actual

- $TP = 21, FN = 6, FP = 7, TN = 41; P = 27, N = 48, \text{total} = 75$
- Accuracy = $\frac{62}{75} = \mathbf{0.827}$
- Sensitivity = $\frac{21}{27} = \mathbf{0.778}$
- Specificity = $\frac{41}{48} = \mathbf{0.854}$
- Precision = $\frac{21}{28} = \mathbf{0.750}$

72 Ch – rows are Predicted

- $TP = 21, FP = 6, FN = 7, TN = 41; P = 28, N = 47, \text{total} = 75$
- Accuracy = $\frac{62}{75} = \mathbf{0.827}$
- Sensitivity = recall = $\frac{21}{28} = \mathbf{0.750}$
- Specificity = $\frac{41}{47} = \mathbf{0.872}$
- Precision = $\frac{21}{27} = \mathbf{0.778}$

Problem 5.5 10**QUESTION AS PRINTED** 2074 Ashwin · Q4 · [10]

The confusion matrix for a classifier is given as follows:

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	25	9
	Class 2	4	31

Calculate: a) Accuracy b) Sensitivity c) Specificity d) Precision

- $TP = 25, FN = 9, FP = 4, TN = 31; P = 34, N = 35$, total 69
- Accuracy = $\frac{56}{69} = \mathbf{0.812}$ Sensitivity = $\frac{25}{34} = \mathbf{0.735}$
- Specificity = $\frac{31}{35} = \mathbf{0.886}$ Precision = $\frac{25}{29} = \mathbf{0.862}$

Problem 5.6 3+5**QUESTION AS PRINTED** 2080 Baishakh · Q3 · [3+5]

In what cases you cannot use Accuracy for performance measure, give some examples. Assume that you have the following confusion matrix. Calculate the Classification error, Sensitivity, False alarm rate Specificity.

		Actual Values	
		True	False
Predicated Values	True	1050	250
	False	150	950

- $TP = 1050, FP = 250, FN = 150, TN = 950; P = 1200, N = 1200$, total 2400
- Classification error = $\frac{250+150}{2400} = \mathbf{0.167}$
- Sensitivity = $\frac{1050}{1200} = \mathbf{0.875}$
- False alarm rate (FPR) = $\frac{250}{1200} = \mathbf{0.208}$
- Specificity = $\frac{950}{1200} = \mathbf{0.792}$ (check: $0.208 + 0.792 = 1 \checkmark$)

Problem 5.7 3+6**QUESTION AS PRINTED** 2073 Shrawan · Q4 · [1+3+6]

Explain ROC. Using the following data, calculate TPR, FPR, precision for given confusion matrix.

	A	B
A	20	5
B	10	40

Classify, A = Yes, B = No

- $TP = 20, FN = 5, FP = 10, TN = 40; P = 25, N = 50$
- TPR = $\frac{20}{25} = \mathbf{0.80}$ FPR = $\frac{10}{50} = \mathbf{0.20}$
- Precision = $\frac{20}{30} = \mathbf{0.667}$
- The ROC point for this classifier is (0.20, 0.80), comfortably above the diagonal.

Problem 5.8 10

QUESTION AS PRINTED 2070 Chaitra · Q3 · [10]

An input sequence “A A B B B A A B B” was used for classification. The Classifier ‘X’ predicted the sequences as: “A A B B B A A B B” where as the Classifier ‘Y’ predicted the sequences as: “A A A A B B A A A B”. Develop the corresponding confusion matrix for the classifiers and find their corresponding.

- i) Accuracy ii) Precision iii) True Positive Rate iv) False Positive Rate

Step 1: tabulate Y position by position

(take A as the positive class)

Position	1	2	3	4	5	6	7	8	9	10
Actual	A	A	B	B	B	A	A	A	B	B
Y	A	A	A	A	B	B	A	A	A	B
Category	TP	TP	FP	FP	TN	FN	TP	TP	FP	TN

Classifier X

(identical to actual)

- $TP = 5, TN = 5, FP = 0, FN = 0$
- Accuracy = **1.0** Precision = **1.0**
- TPR = **1.0** FPR = **0**
- **Perfect classifier** – the ROC point is the top-left corner (0, 1).
- **Reading:** Y has decent recall (0.80) but poor precision (0.571) – it over-predicts A, firing 3 false alarms out of 5 actual Bs. Its ROC point (0.60, 0.80) is only just above the diagonal, so it is barely better than guessing.

Classifier Y

- $TP = 4, FN = 1, FP = 3, TN = 2; P = 5, N = 5$ (total 10 ✓)
- Accuracy = $\frac{4+2}{10} = \mathbf{0.60}$
- Precision = $\frac{4}{7} = \mathbf{0.571}$
- TPR = $\frac{4}{5} = \mathbf{0.80}$
- FPR = $\frac{3}{5} = \mathbf{0.60}$

M6. ANN gradients (backpropagation)**Method**

- Sigmoid and its derivative: $\sigma(z) = \frac{1}{1+e^{-z}}, \quad \sigma'(z) = \sigma(z)(1 - \sigma(z))$.
- Loss $E = \frac{1}{2}(o - t)^2 \rightarrow \frac{\partial E}{\partial o} = (o - t)$.
- Apply the **chain rule** backwards from the loss to the weight in question.

Problem 6.1 2+3+4

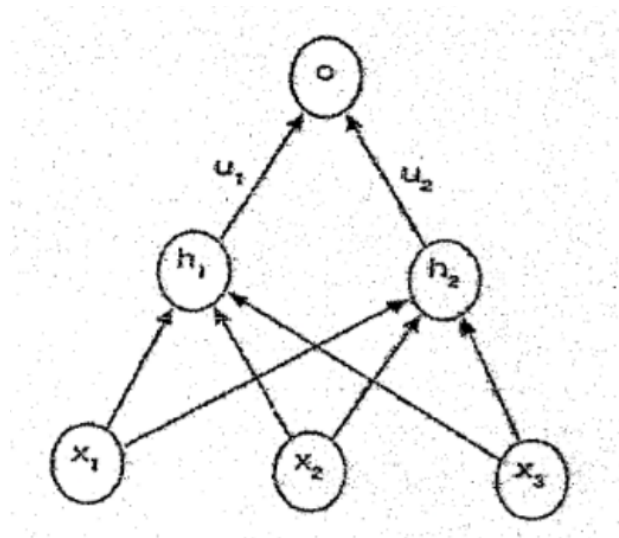
QUESTION AS PRINTED 2076 Chaitra · Q6 · [2+3+4]

Consider the multi-layer feed-forward neural network shown in the following figure. This neural network has three inputs x_1, x_2 and x_3 connected to a hidden layer consisting of two nodes h_1 and h_2 . The weight of the edge connecting x_i to h_j is w_{ji} . The two hidden nodes are connected to the output node o . The weight of the edge connecting the hidden node h_i to the output node o is u_i . The activation functions at hidden and output layers is set to sigmoid function defined as follows:

$$\sigma(\theta) = \frac{1}{1 + \exp(-\theta)}$$

Using the target output t , the squared error is used as the loss function at the output node, and is defined as:

$$E(o, t) = \frac{1}{2}(o - t)^2$$



- Using the symbols given above, compute the activation at h_1 .
- Compute the gradient of the loss with respect to the output o .
- Compute the gradient of the loss with respect to the weight w_{12} .

(a) Activation at h_1

$$h_1 = \sigma(w_{11}x_1 + w_{12}x_2 + w_{13}x_3) = \frac{1}{1 + \exp(-(w_{11}x_1 + w_{12}x_2 + w_{13}x_3))}$$

(b) Gradient of the loss w.r.t. the output o

$$E(o, t) = \frac{1}{2}(o - t)^2 \Rightarrow \frac{\partial E}{\partial o} = (o - t)$$

(c) Gradient of the loss w.r.t. w_{12}

- w_{12} is the weight from x_2 into h_1 , so the chain runs $E \rightarrow o \rightarrow h_1 \rightarrow w_{12}$.
- Write $net_o = u_1h_1 + u_2h_2$ and $net_{h_1} = w_{11}x_1 + w_{12}x_2 + w_{13}x_3$.

$$\frac{\partial E}{\partial w_{12}} = \underbrace{\frac{\partial E}{\partial o}}_{(o-t)} \cdot \underbrace{\frac{\partial o}{\partial net_o}}_{o(1-o)} \cdot \underbrace{\frac{\partial net_o}{\partial h_1}}_{u_1} \cdot \underbrace{\frac{\partial h_1}{\partial net_{h_1}}}_{h_1(1-h_1)} \cdot \underbrace{\frac{\partial net_{h_1}}{\partial w_{12}}}_{x_2}$$

$$\boxed{\frac{\partial E}{\partial w_{12}} = (o - t) o(1 - o) u_1 h_1(1 - h_1) x_2}$$

- Sanity checks:** the factor x_2 must appear (no input, no gradient); u_1 appears and u_2 does not, because w_{12} only influences o through h_1 .
- Weight update:** $w_{12} \leftarrow w_{12} - l \frac{\partial E}{\partial w_{12}}$.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.